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Abstract

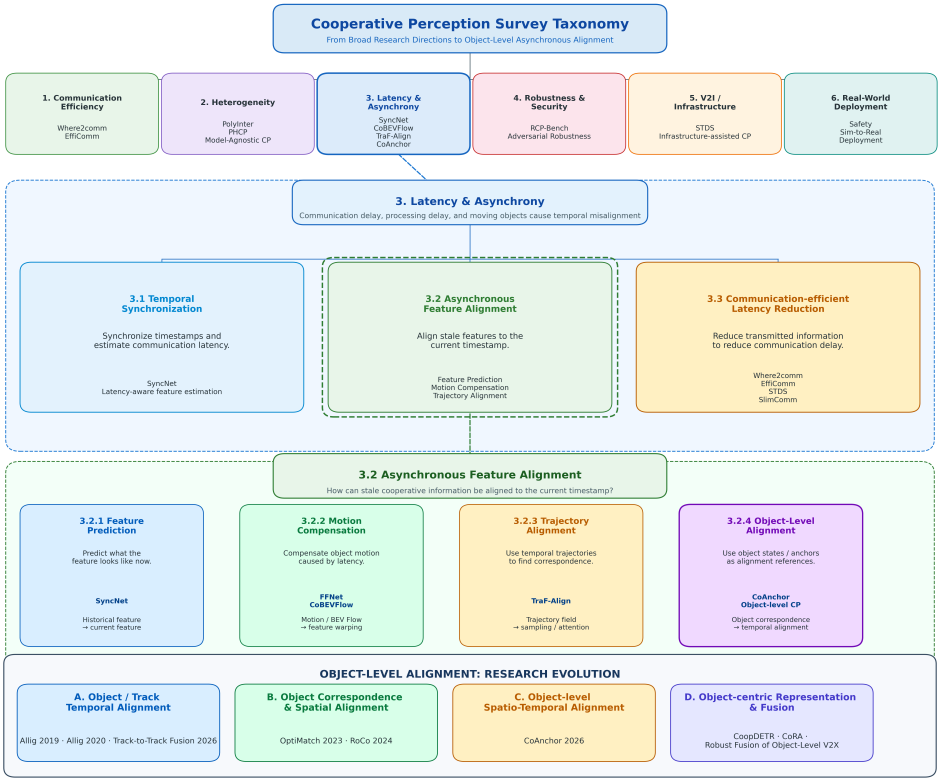


Figure 1: Overview of the survey.

1 Introduction

I had browsed through existing works in recent 3 years in the field of Intelligent Vehicle and summarized the topic and the trend of the research. [1–12]

There are several main problems in the field of Intelligent Vehicle, including bandwidth & communication bottlenecks, latency & temporal asynchrony, heterogeneity & cross-model compatibility, infrastructure-assisted efficiency (V2I), robustness, security & reliability, and sim-to-real gap & empirical deployment constraints.

Then I focus on the problem of latency & temporal asynchrony in cooperative perception. These two challenges are critical to spatial-temporal data consistency. Latency

refers to the delay form sensation to fusion and decision-making, which can be caused by network transmission, data processing, and algorithmic computation. This latency come from bottlenecks in communication, computation, and synchronization, which can lead to outdated or incomplete information being used for decision-making. Temporal asynchrony refers to the misalignment of data timestamps from different sources, including RSUs and CAVs, which can lead to inconsistencies in the perception results.

Usually, data are fused trivially by addition or concatenation, which can lead to spatiotemporal misalignment and inaccurate perception results. Moving objects are predicted to be in wrong historical positions instead of their current positions, leading to phantom detections, tracking fragmentation, and dangerous miscalculations in downstream motion planning.

And nowadays, there are several methods to address the latency and temporal asynchrony problem in cooperative perception, each compensating from different aspects. [13–18]

These include predicting current features using historical information, performing spatial warping of features via motion information or BEV flow, and leveraging trajectory information to establish correspondences across different timestamps. However, these methods align features in a single frame, which may not fully capture the temporal dynamics of moving objects and may not be robust to occlusions or missing data. Therefore, recently research moves toward object level.

2 Object level Alignment

Core concept of Object-Level Alignment is to use the object itself as a correspondence reference between different agents and different timestamps. For example, object state, bounding box, track, trajectory, or object-level anchor can be used as the reference. Compared to directly performing temporal compensation on the entire feature map, object-level methods can estimate the position, velocity, and temporal changes of individual objects, and then propagate historical object information to the current timestamp. Asynchronous feature alignment shifts temporal compensation from the feature level to an object-centric paradigm. By leveraging more precise object correspondences and motion cues, it can mitigate the adverse effects of temporal asynchrony on cooperative perception.

2.1 Existing Object-Level Alignment Approaches

Existing object-level approaches can be grouped according to how they address temporal inconsistency, cross-agent correspondence, and spatial misalignment. The first direction focuses on object or track temporal alignment, where object states are predicted or fused across different timestamps to handle asynchronous observations. For example, track-to-track fusion methods explicitly consider asynchronously arriving object tracks and investigate different state-fusion strategies for maintaining a consistent global object representation [**?**].

The second direction focuses on object correspondence and spatial alignment. RoCo formulates pose correction as an iterative object-matching and graph-optimization problem, refining agent poses based on the alignment errors of matched objects [**?**].

More recently, CoAnchor addresses the joint challenge of communication delay and relative-pose noise. Instead of directly reasoning over dense bird's-eye-view features, it

introduces sparse object-level spatio-temporal anchors as a shared interface for spatial refinement, temporal propagation, and current-time verification [18]. This provides a useful case study for examining how object-level representations can connect temporal compensation and spatial correction within a unified collaborative perception framework.

In the following, we take CoAnchor as a representative example of integrated object-level spatio-temporal alignment and examine its problem formulation, design choices, technical components, and experimental evidence.

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